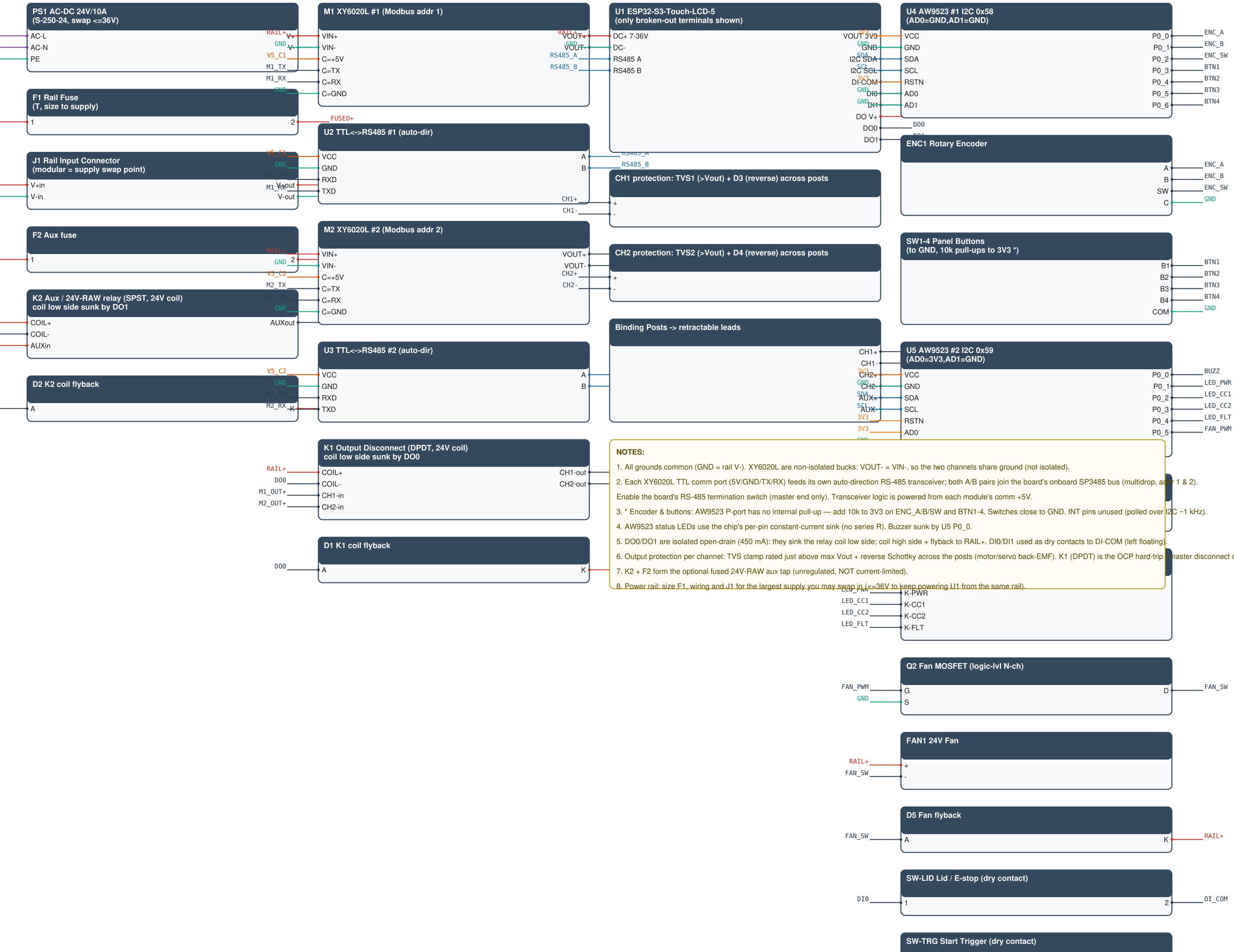
Dual-Channel Bench Supply — System Schematic

ESP32-S3-Touch-LCD-5 · 2× XY6020L on RS-485 Modbus · 2× AW9523 I²C · relays on DO0/DO1 · switches on DI0/DI1 · 24 V rail

Net-labelled schematic (labels of same name are connected). Rev A · 2026-06-23 · generated



NOTES:

1. All grounds common (GND = rail V-). XY6020L are non-isolated bucks: VOUT- = VIN-, so the two channels share ground (not isolated).
2. Each XY6020L TTL comm port (5V/GND/TX/RX) feeds its own auto-direction RS-485 transceiver; both A/B pairs join the board's onboard SP3485 bus (multidrop, addr 1 & 2). Enable the board's RS-485 termination switch (master end only). Transceiver logic is powered from each module's comm +5V.
3. * Encoder & buttons: AW9523 P-port has no internal pull-up — add 10k to 3V3 on ENC_A/B/SW and BTN1-4. Switches close to GND. INT pins unused (polled over I2C ~1 kHz).
4. AW9523 status LEDs use the chip's per-pin constant-current sink (no series R). Buzzer sunk by U5 P0_0.
5. DO0/DO1 are isolated open-drain (450 mA): they sink the relay coil low side; coil high side + flyback to RAIL+. DI0/DI1 used as dry contacts to DI-COM (left floating).
6. Output protection per channel: TVS clamp rated just above max Vout + reverse Schottky across the posts (motor/servo back-EMF). K1 (DPDT) is the OCP hard-trip master disconnect on both + leads.
7. K2 + F2 form the optional fused 24V-RAW aux tap (unregulated, NOT current-limited).
8. Power rail: size F1, wiring and J1 for the largest supply you may swap in (<=36V to keep powering U1 from the same rail).